



Integrating Marketing, Operations, and Purchasing to Create Value

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For years many researchers and practitioners have been calling for a more boundary-spanning or cross-functional view of the management of businesses. As any student of optimization will note, the combination of several optimized sub-systems will rarely, if ever, result in a globally optimized system. It is much the same with a business. Decisions that may appear extremely good within the confines of a single organizational area are often clearly dysfunctional when viewed from other areas of the organization. To effectively compete in today's marketplace, firms need to effectively link their diverse functional areas. This article provides an example of how the marketing, operations, and purchasing areas can be effectively linked to accomplish the common goal of effectively competing in the value market segment of consumer goods and services.

Key words—decision support systems, effectiveness, group decisions, integration, retailing

INTRODUCTION

IN THE USA AND VARIOUS other countries, there has been a significant shift in consumer tastes toward more value-oriented products and services, where value is often loosely defined as a perceived quality-to-cost ratio. While the value segment has long been the dominant sector in consumer goods and services, this segment has grown significantly in recent years. Stagnant incomes, a global recession, and attitudinal changes to more conservative lifestyles have all contributed to the tremendous growth of the value segment for consumer products and services. In response to the growth of the value segment, many firms are now rushing to re-position themselves as providing value products and services.

The expected result has been an increase in competition in this already highly-competitive segment of the market. This increased competition has strained the ability of firms worldwide to remain profitable. These firms must often reduce their prices to improve the per-

ceived value of their products. The natural result is increased pressure on firms to control costs.

The purpose of this paper is to identify and demonstrate how an integrated approach to marketing, operations, and purchasing can provide opportunities for a firm to lower costs and increase sales of products and services in the value market segment. Integration of these diverse functional areas allows a retail firm to effectively link the supplier negotiation and selection tasks, as well as the buying decisions of the purchasing area; the store layout and management decisions of the operations area; and the merchandising decisions of the marketing area. The retail setting used to illustrate this integrated approach is the grocery industry. The grocery industry provides an excellent backdrop because it represents a crowded and very large retail market with standard and substitutable products, highly visible dominant players (supermarkets), a large value market segment, and extreme cost competition. Benefits from better integrating these separate functional

areas in the grocery industry setting described in this paper include, but are not limited to, significantly lower procurement costs, an increased cash float position, a more profitable sales mix, and an increase in the overall sales of the firm. Although the benefits of integration described in this paper can be successfully applied in the value segment of most retail environments, the grocery industry provides a specific application where the results can be especially dramatic.

The remainder of the paper is organized into five sections. A review of related literature is provided in the first section. In the second section the competitive environment of the value segment of the grocery industry is overviewed. The core elements of an integrated approach to marketing, operations, and purchasing are identified and then demonstrated in the third and fourth sections. Finally, conclusions are drawn in the last section.

A BRIEF REVIEW OF RELATED LITERATURE

Numerous authors have identified cross-functional cooperation and boundary-spanning decision processes as fertile ground for improving overall business performance. Marketing researchers have for years noted the important roles that other functional areas play in the development and implementation of a successful marketing plan [15, 34]. Studies have addressed the integration of marketing with finance [2], research and development [12], operations [22], production planning [31], purchasing [16], and other areas of the organization [28]. Similarly, purchasing researchers have for a long time cited the benefits of linking purchasing decisions to other functional areas, as well as a firm's strategic decision process. [10, 27, 30, 33]. Mathematical and conceptual models have been proposed to provide a more integrated decision process between diverse functional areas (see [12, 23, 31] for a few examples). Porter [24] has noted that firms can gain competitive advantage by effectively managing these linkages. Nonetheless, it is readily acknowledged that the vast majority of firms have not implemented boundary-spanning decision processes [1, 10].

Most boundary-spanning research studies have focused on the strategic benefits of cross-functional linkages. Firms need to have sufficient production capabilities to meet marketing projections, research and development efforts that are linked to marketing plans, financial plans that facilitate investment in necessary production facilities, etc. However, recently researchers have begun to take a closer look at tactical and operational linkages among functional areas. Mahajan *et al.* [22] have identified service operations that involve task interdependence among workers in separate functional areas. They note that functional roles often are so interwoven in service firms that individual roles can become unclear. In many situations it may be beneficial for service providers (the operations function) to be the same people that market the service. Thus, there exist numerous scenarios where decision processes are best linked at all levels in the organization.

This research extends the existing literature in this area by examining an integrated decision process among multiple functional areas (three, in this case), extending linkages to the operational or task level, and providing a specific demonstration of how a cross-functional approach to managing a retail firm can be used to create competitive advantage. As such, this article can be viewed as a demonstration of what Hammer and Champy [14] have recently coined as the 'reengineering' of firms. In the example presented here, integration (or reengineering) of the marketing, operations, and purchasing processes is shown to create value for retail grocery firms.

COMPETITION FOR THE VALUE SEGMENT IN THE GROCERY INDUSTRY

The grocery industry in the USA and most other industrialized nations is extremely well defined with numerous major players and extreme cost competition. In the USA, approximately three-quarters of annual grocery sales (estimated to exceed \$400 billion) are incurred by supermarkets, which are defined as full-line, full-service food stores with annual sales above \$2 million. Most of these supermarkets are operated by chain or independent owners that typically achieve rather small gross and net margins, estimated to be around 22 and 2%, respectively.¹

¹Data extracted from *Progressive Grocer's 59th Annual Report of the Grocery Industry*, April 1992.

These small grocery margins are the result of a shift in consumer preferences to more value-oriented products and the inability of most grocery firms to successfully differentiate their products. A recent consumer survey showed 76% of those surveyed regard low prices as extremely important in rating grocery stores, second only to cleanliness at 77%. In addition, this survey reported that 72% of grocery consumers shop in more than one supermarket each week, with the primary store receiving 74% of their food dollars (see footnote 1 on p. 160). These results are no surprise to industry professionals who have witnessed a steady decline in both brand and store (or chain) loyalty. Volumes have been written in the recent business press about the difficulty in attracting and maintaining consumer loyalty in the grocery industry, as well as other retail businesses (see [8, 11, 21, 32] for a small sample). Differentiation is particularly difficult for grocery retailers to achieve because there are numerous standard grocery products that all supermarkets must offer. With the exception of private-label products, non-price advertising of products is for the most part outside the control of individual grocery firms and does not provide competitive advantage to any one firm.

Often the most successful grocery firms are those that can provide products at lowest cost to the value-oriented consuming public. This is a particularly difficult task in the grocery industry where competing firms are of similar size, have similar cost structures, have access to the same vendor deals, and have good information concerning competitors' prices and merchandising strategies. In the USA these conditions exist in almost all grocery markets, even those that are extremely small. Under these conditions, competition in local grocery markets often becomes a zero-sum game where one firm's gains are another's loss. In this type of environment it becomes increasingly more difficult to gain competitive advantage. Conventional marketing, operations, and purchasing plans can be easily matched by competitors. What is needed is a plan that integrates these areas and exposes potential plans that cannot be obtained when these areas are operated separately, as is done in most firms.

INTEGRATING MARKETING, OPERATIONS, AND PURCHASING

From a purely mathematical viewpoint, it may be beneficial to integrate several business

areas and search for globally optimal solutions rather than implement several separate locally optimal solutions. The marketing, operations, and purchasing areas in grocery firms represent an example where integration can lead to tactical plans that cannot be effectively matched by competitors and can improve a firm's profitability. In a conventional organization, the purchasing area negotiates with vendors and purchases product; the marketing area promotes and merchandises these products to the consuming public; and the operations area manages the delivery and transfer mechanisms. Even though there exists a significant amount of interaction between these areas (test marketing of new products being one example where purchasing and marketing personnel interact), the vast majority of grocery firms do not fully integrate these areas in an effort to seek competitive advantage. The following example demonstrates the role that each of these areas can play in an effectively integrated tactical plan for generating competitive advantage.

The role of purchasing

In conventional organizations the purchasing area is designed to effectively process and track product orders. The primary responsibilities of the purchasing area are to obtain the lowest price possible from vendors and maintain an uninterrupted supply of product to ensure the success of the marketing and operations plans. In the grocery industry this usually results in extremely large buys from vendors that offer attractive promotional discounts. These discounts are offered in regional markets by vendors in an attempt to improve their competitive positions in those markets or accelerate sluggish sales. Purchasing personnel in conventional grocery firms regularly sift through these offers and purchase extremely large quantities of those products they feel they can sell.

There are two basic flaws with this conventional approach to the purchasing area. First, the linkages to other segments of the organization are relatively loose. Although the goal of purchasing is to effectively service these areas, there are no mechanisms to closely integrate the decision processes with these other areas. Opportunities for developing better global solutions are missed. Secondly, huge amounts of cash are needed to purchase large quantities of discounted product. As a result, the purchasing

area develops into a cash user rather than a cash generator. An integrated approach can help correct these flaws.

Consider first the process of dealing with vendors. As noted earlier, the cost of goods sold in the grocery industry averages 78¢ for every dollar of sales. Thus, an attractive alternative for improving the profitability of a grocery firm is to reduce procurement costs. In an integrated approach, the purchasing area constantly works with the marketing and operations areas, as well as vendors, to develop new ways for lowering the firm's average cost of goods sold. By offering promotional discounts, vendors have already shown their willingness to participate in this process.² What is needed is an integrated approach to shape this willingness into creative plans that can enhance the financial performance of both the grocery firm and individual vendors.

One way of doing this is to create opportunities for vendors with slow selling brands in each of the product categories in a grocery firm. For the purposes of this research a grocery firm's products are subdivided into a thousand or so categories of substitutable products. For instance, sales of 16 oz. cans of sliced yellow cling peaches in heavy syrup may represent a product category comprised of national brands such as DelMonte and Libby's, a private-label brand and maybe a regional brand. Usually, the vendors with the number one selling products in a market employ a distributive (win-lose) negotiation process. They dictate the deal and the grocery firms take it. However, other vendors with less desirable market positions (often not even on the shelf) are likely to be more amenable to a more integrative (win-win) negotiation process.³ These vendors (and their products) represent excellent opportunities that are often overlooked in conventional organizations.

Consider the plight of a vendor with a brand performing extremely poorly in a local market. To increase their share in this market they need to heavily promote their product. Unfortunately, few grocery firms are willing to consistently participate in their promotions because

their product is such a poor performer. To add insult to injury, their product may actually be of rather high quality and may even be performing extremely well in other markets. This vendor will very likely provide an excellent discount and/or extremely favorable terms to a grocery firm in exchange for a long-term promotion of the vendor's product. Because the vendor can divert promotional money from other markets where their product is performing well and since it is unlikely that other grocery firms in this market will participate in this promotion (insufficient sales volume), the vendor can agree to even a sweeter deal than they would have otherwise. The vendor's hope is that high sales of their product at one grocery firm will motivate other firms in the market to participate in future promotions of their product. The gain of the negotiating grocery firm is that it has created an opportunity to create value by reducing procurement costs and providing the marketing area with a tool for beneficially altering the mix of category sales, as will be discussed later. An integrated approach toward purchasing helps to identify these opportunities.

Secondly, the purchasing area needs to become a source of capital rather than a drain on the firm's cash. This can be done by extracting terms and conditions from vendors attempting to improve their market positions that favorably improve the grocery firm's cash float position. These vendors will be more amenable to providing attractive terms because of their weak market positions. Grocery firms that understand the value of cash float can make intelligent purchasing decisions and more fully evaluate the profitability of competing products.

Evaluating the effect that negotiated terms have on the cash flows of a firm is nothing new. As one would expect, the financial literature is rich with detailed cash flow analyses for different payment and receipt schedules (see [3] for several examples). In addition, numerous studies have evaluated the effects of opportunity costs, inflation, and payment schedules on optimal order quantities [5, 6, 7, 9, 13, 25, 26, 29]. However, none of these previous studies has investigated how negotiated terms can be linked to an integrated decision process, which is the primary focus here. The following analysis will serve to quantify the value of float and show how it can be exploited to improve the profitability of a grocery firm.

²See [17] for an overview of typical grocery promotional deals and [4] for antitrust and legal limitations on promotional deals in the USA.

³As defined by Lewicki and Litterer [19], an integrative negotiation process provides an opportunity for both parties to benefit by working together.

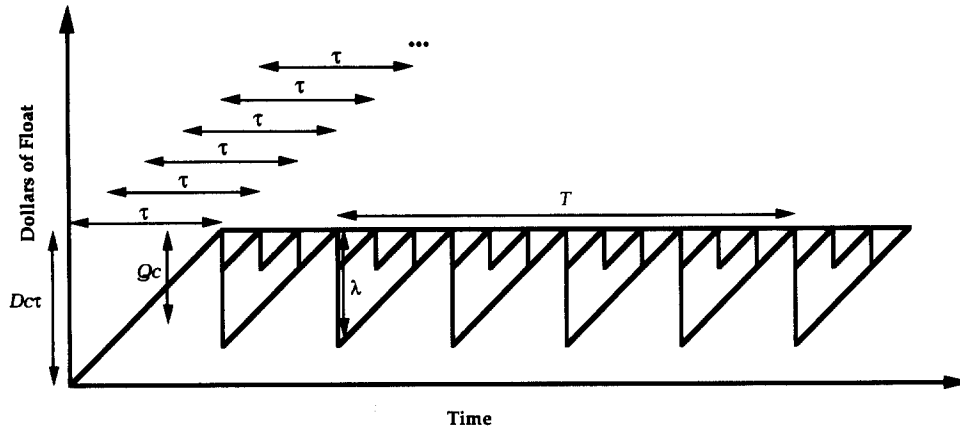


Fig. 1. Cash float.

The level of a firm's cash float over time for a single product is depicted in Fig. 1. For the purposes of this research, cash float is defined as the funds that are available to a firm because of a lag in time between when purchased goods are received and when vendors are paid. If payment for an order is due τ days after receipt of Q units and the price and cost of each unit are represented by p and c , respectively, then after τ days of D units of daily demand⁴ the non-compounded value of this purchase will be equal to $Dp\tau$ dollars. From this total, $D(p - c)\tau$ represents the gross profit from selling the product and $Dc\tau$ represents cash float. After τ days, a payment equal to Qc must be made to the vendor. This payment, which must be made τ days from each order, represents a reduction in cash float from the maximum attainable level ($Dc\tau$). The interested reader can find the derivation of a mathematical expression for cash float, as well as contribution from sales and profits, in Appendix I.

In addition to maximizing the product's contribution cash stream, purchasing should also attempt to maximize cash float. The rather obvious mechanisms for doing this are to increase the repayment period (τ) and reduce order quantities (Q). These mechanisms can be integrated into the vendor negotiation process. Vendors should be managed in a way that maximizes both the profits from sales and the value of the cash float from the product. Cash float has considerable value to a firm because it can be used to pay operating expenses and/or it

can be invested in projects. In some cases the selection of which brand to promote to improve a product category may be determined by the amount of float created in addition to the negotiated procurement cost.

Consider the following example with two competing brands.

Brand	D (units/day)	Q (units)	τ (days)	p (\$)	c (\$)	Net margin
1	200	5000	15	1.00	0.78	2%
2	200	1000	45	1.00	0.79	1%

The negotiated procurement cost for brand 1 yields typical grocery gross and net margins of 22 and 2%, respectively. Although the net margin for brand 2 is one-half that of brand 1, better float terms have been negotiated for this item. If the expected annual return from investments (i) is 29% for this firm, the profitability of these two products over one year [equations (4)–(6) in Appendix I] would appear as follows:

Brand	Contribution from sales (\$)	Profits from sales (\$)	Profits from investing float (\$)	Total profits (\$)
1	18,241	1658	112	1770
2	17,412	829	1949	2778

In this case, even though brand 1 generates twice the net profits from sales of brand 2, brand 2 is more profitable overall because of the value of the cash float it generates for the firm. If a promotion was planned for one of these two competing brands, conventional wisdom would suggest selecting brand 1 because of its higher net margin. However, when the value of cash

⁴For convenience demand is assumed to be constant. In Appendix III it is shown that this assumption does not appreciably affect the analyses presented in this article.

float is included in the analysis brand 2 is shown to be significantly more profitable. In this case, consideration of the value of cash float has altered a promotional decision.

The role of marketing

An integrative negotiation process and an analytical representation of cash float can provide numerous opportunities if effectively integrated with the efforts of the marketing area. As part of the negotiation process the managers of the grocery firm may agree to heavily promote a vendor's product on a long-term basis (possibly by prominently displaying it in their weekly flyer). The goal of this marketing effort is to cannibalize sales from other products within the category (as well as expand the category sales) to improve the profitability of the category. A mathematical relationship describing the benefits from changing a category's sales mix is provided in Appendix II.

The repayment periods, delivery quantities, unit purchase costs, and selling price discount [parameters τ_j , Q_j , c_j ($j = 1, \dots, n$), and α respectively in equations (7)–(11) in Appendix II] represent outputs from the negotiation process with vendors. Under normal circumstances negotiations take place with one vendor at a time so that only τ_1 , Q_1 , c_1 , and α are outputs that may vary. The goal of grocery managers is to negotiate a set of outputs that improve the bottom-line performance of the firm. Equations (7)–(11), which demonstrate the effects from increased revenue, as well as increased cash float and cannibalization, can be used to evaluate new vendor proposals or seek out opportunities for improving the performance of the category. Historical performance and/or managerial intuition can be used to determine the cannibalization rates ($\phi_j, j = 2, \dots, n$) that can be achieved for different values of α . Alternatively, the negotiation process can be evaluated by determining sets of ϕ_j breakeven values for particular negotiations. Although estimating the ϕ_j variables appears to be a rather complex undertaking, in most situations it is actually fairly straightforward. Most product categories are limited to 4 or fewer products with individual product sales that tend to follow a Pareto distribution. In many situations, several products with lower sales can be combined into a single alternative product without affecting the general conclusions of the analysis. As a result it is seldom,

if ever, necessary to estimate more than one or two cannibalization factors (ϕ_j). Furthermore, the ϕ_j values are highly correlated to α and most grocery managers can quickly determine if the computed ϕ_j values are within the realm of the possible. As a result, equations (7)–(11) in Appendix II can be effectively used to integrate the negotiation tasks of purchasing to the merchandising tasks of the marketing area.

This integrative process provides marketing with a tool that allows them to reduce prices and improve overall margins; an important combination in the value segment. Because of successful negotiations from the purchasing area and well thought-out store operations to be discussed next, marketing can focus on directing consumers to value products that enhance the firm's competitive position. Not only can this approach help to modify the buying habits of current customers, but it can also enhance the prospects for additional category sales from new customers. These new customers may decide to pick-up some of these 'bargains', even though their primary store may be a competitor's. Possibly these new customers may buy additional items, or may even switch primary stores.

The role of operations

Now comes operation's role in the integrated decision process. The relative profitability of products and strategies for cannibalizing sales of less profitable products needs to be communicated to operations so that they can provide an effective link between the firm's purchasing and marketing efforts. The layout of stores needs to be designed not only to communicate the firm's value image and keep overheads relatively low, but also to facilitate the reshaping process. The high-value alternative products need to be clearly marked and conveniently located so that consumers can make decisions that improve the mix of category sales. Possibly even more importantly, store personnel need to be aware of and motivated toward successfully implementing this process. Displays need to be well stocked and maintained attractively to be successful. It is not unusual for store personnel to unknowingly defeat marketing campaigns by purposely 'hiding' sale items that they suspect are loss leaders in order to "improve" the company's profitability. A reward system has to be initiated that links the success of revising the

sales mix at each store (or ultimately, each department or product category) to the compensation of the personnel at that location.

The development of both formal and informal lines of communication may be instrumental in enabling store operations to effectively implement these sales mix changes. For instance, an information system could be developed based on store scanners that provides timely feedback (say weekly) on store and department performance. In addition to monetary rewards for excellent performance, friendly competitions could be developed within and between stores to further motivate the process. Bonuses could be rewarded for imaginative ideas (layout improvements, display adjustments, . . .) for improving the profitability of the sales mix and particularly successful techniques could be communicated to other stores and departments to contribute to the organizational learning process. In this manner, an integrative approach can be significantly enhanced by the management of store operations.

AN EXAMPLE OF THE BENEFITS OF INTEGRATION

Perhaps the best way to show how the integration of the marketing, operations, and purchasing areas can provide opportunities for improving a firm's bottom-line performance that otherwise would not be possible is to look at a small example for a single product. To do this we return to a simple nondescript grocery product category: 16 oz. cans of sliced yellow cling peaches in heavy syrup. Assume that a grocery firm sells three competing brands of this product under the following conditions:⁵

Brand (<i>j</i>)	D_j (cases/day)	Q_j (cases)	τ_j (days)	p_j (\$)	c_j (\$)
1	50	2000	15	33.36	26.00
2	250	2000	15	33.36	26.00
3	40	2000	15	34.80	27.10

Defining, as before, the expected annual return from investments for this firm to be 29%, the current contribution from this product

category over one year's time will be equal to the following [see equation (7) in Appendix II]:

Brand (<i>j</i>)	Contribution from sales (δ_s)	Profits from investing float (π_f)	Total contribution from sales and float ($\delta_s + \pi_f$) _{<i>b</i>}
1	\$152,564	-\$1922	\$150,642
2	\$762,821	\$20,743	\$783,564
3	\$127,690	-\$3192	\$124,498
Totals	\$1,043,075	\$15,629	\$1,058,704

Now let's say that the purchasing area engages in preliminary negotiations for the following new terms, which include the consolidation of deliveries with other products, from the brand 1 vendor:

Brand (<i>j</i>)	Q_j (cases)	τ_j (days)	c_j (\$)
1	500	15	18.50

Normally these newly negotiated terms will include certain volume, promotion, and/or retail price restrictions to improve the vendor's market position. Suppose that these restrictions result in a 20% discount in the retail price of brand 1 ($\alpha = 0.20$) and the managers of the grocery firm feel they will be able to cannibalize a minimum of 50% of the sales of brands 2 ($\phi_2 = 0.50$) and 3 ($\phi_3 = 0.50$) at this discounted price, which will allow the grocer to meet the vendor's required contract volume. The negotiation of these attractive terms by purchasing and the implementation of an effective plan by marketing and store operations (which was a key element in the negotiation process) can directly improve the performance of this product category. Perhaps even more importantly though, this newly modified product category provides additional opportunities for marketing and operations to demonstrate to consumers the value of the firm's products. Heavy promotion of brand 1 can be used to motivate shoppers who normally buy their canned peaches elsewhere to buy their canned peaches (as well as other items) during a quick visit to this firm's stores. This can serve to emphasize the firm's value image, as well as increase the sales of highly-profitable products. Suppose that heavy promotion of brand 1 is projected to increase the sales of the product category by 20% ($\gamma = 0.20$). If these estimates

⁵Although this example is loosely based on experience with an east coast US grocery firm with approx. 200 retail outlets, the actual values and outcomes reported here are for demonstration purposes only.

are accurate, this product category will appear as follows after these changes:

Brand (<i>j</i>)	β_j (cases/day)	Q_j (cases)	τ_j (days)	c_j \$
1	263	500	15	18.50
2	125	2000	15	26.00
3	20	2000	15	27.10

Using equations (8)–(10) in Appendix II, the total contribution from this product category will now appear as follows:

Brand (<i>j</i>)	Contribution from sales (δ_s)	Profits from investing float (π_f)	Total contribution from sales and float ($\delta_s + \pi_f$) _a
1	\$892,767	\$19,837	\$912,604
2	\$381,410	\$6588	\$387,998
3	\$63,845	–\$5597	\$58,248
Totals	\$1,338,022	\$20,828	\$1,358,850

The net result of this integration of marketing, operations, and purchasing tasks would be a significant improvement in the profitability of this example product category ($\Delta =$

\$1,358,850 – \$1,058,704 = \$300,146). The contribution from sales will increase, as will the value of the firm's cash float. Value will be added by reducing the procurement cost of brand 1, increasing the overall cash float position of the product category, cannibalizing sales of brands 2 and 3, and increasing the sales of the product category.

Now suppose that the managers for the brand 1 vendor are vacillating on the \$18.50 procurement cost. Instead of the \$18.50 per case cost they propose a \$19.00 procurement cost, with all other negotiable conditions remaining the same. Figure 2 shows the financial benefit (Δ) to the grocery firm under the initial conditions (point A) and under these new conditions denoted as point B (\$246,166). Rather than accept this significant reduction in financial benefit, the managers at the grocery firm may try to "repackage" the overall program by "splitting the difference" between the two proposed costs (\$18.75) and extending the repayment terms to 60 days (point C). This will actually increase the financial benefit for the grocery firm to

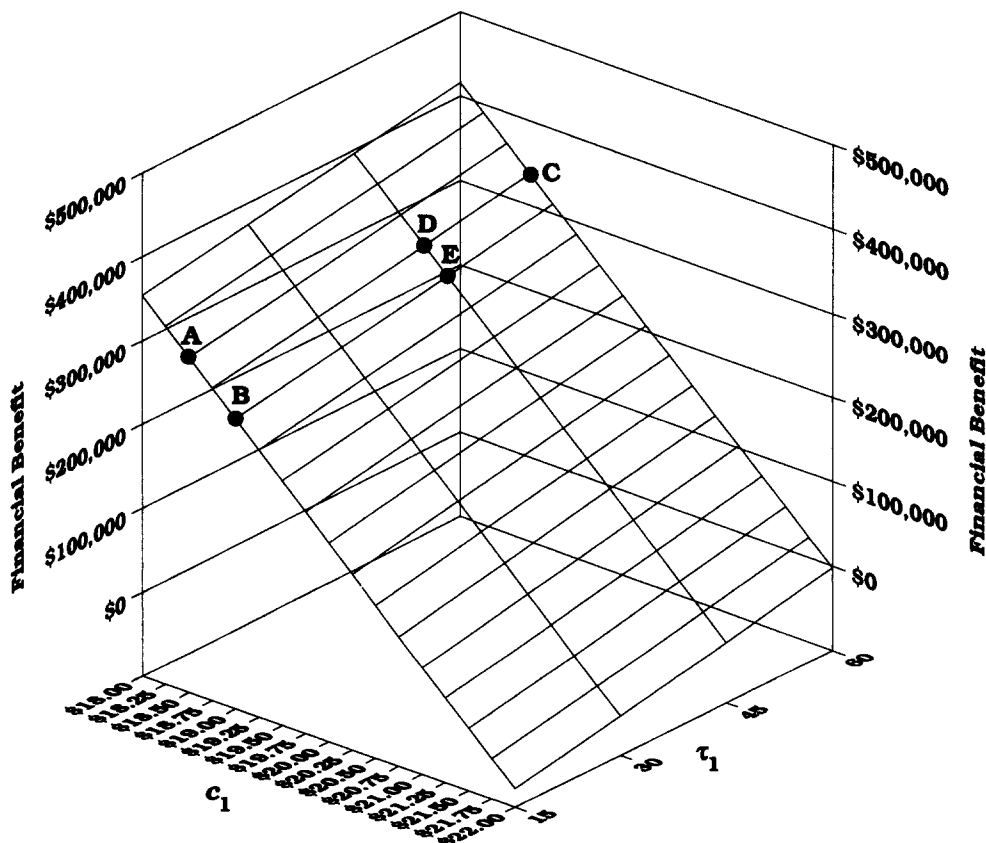


Fig. 2. Trade-off on purchase cost and repayment terms for example scenario.

\$337,555. Ultimately, a compromise may be reached around points D ($c_1 = \$18.75$, $\tau_1 = 45$, $\Delta = \$316,089$) or E ($c_1 = \19.00, $\tau_1 = 45$, $\Delta = \$289,671$).

Suppose that the negotiation with the brand 1 vendor leads to position E. This will result in the following projected total contribution from this product category.

Brand (j)	Contribution from sales (δ_s)	Profits from investing float (π_f)	Total contribution from sales and float ($\delta_s + \pi_f$) _a
1	\$838,250	\$63,878	\$902,128
2	\$381,410	\$6588	\$387,998
3	\$63,845	-\$5597	\$58,248
Totals	\$1,283,505	\$64,869	\$1,348,374

This position allows the managers of the grocery firm to recapture 81% of the potential reduction in the financial benefit from the increase in procurement cost to \$19.00 per case. This higher procurement cost is offset by a significant increase in the value of cash float.

This simple example demonstrates how an integrative decision process can be used to enhance a firm's value image and improve the firm's financial performance. Here a profitable, but under-performing, product category was significantly improved in a manner that would not have been possible without integrating the marketing, operations, and purchasing areas. Under normal circumstances, the purchasing area would not consider negotiating a contract with a vendor that dictated significantly higher sales volume than currently possible. Similarly, marketing and store operations would not consider a major promotional effort for an already profitable product category, especially one that would very likely cannibalize sales from other profitable products. In traditional organizational structures the negotiation process (where the procurement cost, c_1 repayment period, τ_1 , and order quantities, Q_1 for discounted brand 1 are determined) is located in the purchasing area, while the required expertise for evaluating the effect of promotional efforts on product sales (defined by the discount factor, α , the category growth factor, γ , and the cannibalization proportions, ϕ_j) is located in the marketing and operations areas. Separation of these processes does not allow a firm to evaluate the numerous trade-offs necessary for identify-

ing and exploiting new opportunities for enhancing the firm's value image and/or improving the firm's financial performance.

Even though it is unlikely that all of the several thousand product categories can be significantly improved by taking a more integrative approach, there certainly are numerous untapped opportunities. The current industry trend toward reducing the number of product offerings in each product category has put significant pressure on vendors to reposition their products to ensure future market access. Many quality brands have recently been dropped by large chain stores. These and other brands with weak sales in regional markets represent excellent opportunities for the integrative approach described in this article. As other grocery firms continue to drop quality brands with weak sales, the opportunities will continue to increase. Interestingly, this trend also enhances this approach by increasing the likelihood that competing firms will turn down the new promotional programs negotiated with these vendors because these grocery firms no longer carry these products. And finally, consumers that prefer these slow-selling brands that have been discontinued at competing stores will have another reason to visit the integrated grocery firm's stores. Although slow-selling brands (before being discounted and heavily promoted) may represent a small percentage of total sales, a large percentage of the grocery-consuming public will buy at least one of these products on a regular basis. Offering many of these discontinued brands at extremely attractive prices will help a grocery firm's marketing program reach a larger audience.

This integrative decision process is especially powerful because; not only does it link the efforts of the marketing, operations, and purchasing areas; but it also creates an integrative (win-win-win) relationship with vendors and consumers. This process provides vendors that have poorly performing or discontinued products an important strategic link in their effort to reestablish their products in markets, as well as develop a value positioning strategy for their products. This 'back to the wall' situation can provide the impetus for many exciting integrative plans that would not have otherwise been considered. Consumers are provided with an opportunity to purchase these under-performing brands at very attractive prices. As a result,

many of the organizational tasks within the grocery firm that were previously perceived as unpleasant zero-sum political battles or mundane repetitive tasks can now become exciting opportunities for improving the firm's financial performance and significantly enhancing the firm's value image. Everyone wants to win and an integrative approach provides a mechanism for all participants to win. Opportunities become available for a grocery firm to profitably enhance their competitive environment, a vendor to improve the competitive position of an under-performing product, and customers to purchase alternative brands at significantly lower prices. An integrative decision process creates a win-win-win linkage of a grocer, vendors, and the grocery-consuming public.

CONCLUSIONS

A shift in consumer tastes has made the value segment of many retail industries a highly sought after and hotly contested market. The natural result has been increased pressure on firms to decrease their costs (prices) to improve consumer perceptions of their relative value and improve their market position. Firms that are successful in providing consumers with lower prices for similar (or the same) products can create competitive advantage. This paper shows how a more integrated decision process can enable firms to accomplish this goal. The grocery industry example used here demonstrates the potentially dramatic benefits of integration.

The contributions of this paper are threefold. First, a specific approach for improving the bottom line performance of retail firms by integrating the decision process is demonstrated. Many researchers have argued that firms should develop a more integrated approach to managing their businesses, but the details of how this should be done at the tactical and operational levels have been sparse. While the benefits of integration go well beyond the demonstration provided in this article, the approach for integrating marketing, operations, and purchasing described in this article represent tangible evidence of the benefits of integration.

The second and third contributions of the paper are the analytical representations of the value of cash float and benefits from altering a

firm's sales mix. These analytical tools can be used within the vendor negotiation process; to evaluate alternative vendor proposals; measure the profitability of products; or evaluate marketing, operations, and purchasing plans. The potential benefits (or losses) from implementing an integrated plan are analytically estimated by these representations. The result, is a more structured analysis of the integrated decision process.

As this paper has demonstrated, integration of the marketing, operations, and purchasing areas can help managers of a firm make business decisions that result in better performance than would otherwise be possible if these areas continued to operate separately. These integrated decisions can potentially create value for the firm, its suppliers, and its customers. From an operations research perspective, the firm's managers extend the solution space beyond their previous concentration on local optima in search of better global decisions. In many retail scenarios, such as the grocery industry, this integrated approach can result in a significant improvement in overall performance.

APPENDIX I

Determining Mathematical Expressions for Cash Float Contribution, and Profits

Referring to Fig. 1, if payment is due τ days after receipt of Q units each with a purchase cost equal to c , then after τ days of D units of daily demand the non-compounded value of cash float created will be equal to $Dc\tau$. After τ days, a payment equal to Qc must be made to the vendor. This payment, which must be made τ days from each order, reduces cash float from the maximum attainable level ($Dc\tau$). If the per day opportunity cost of capital equals i , then the current value of the difference between the maximum level of cash float and the actual level over each order cycle is equal to λ , where

$$\begin{aligned}\lambda &= \int_0^{Q/D} [(-Qc + Dct)i] \exp(-it) dt \\ &= -Qc + Dc \left[\frac{(1 - \exp[-i(Q/D)])}{i} \right]. \quad (1)\end{aligned}$$

If all order quantities are equal to Q , then the current value of the reduction in the maximum attainable cash float can be represented by an

infinite geometric series (θ), where

$$\begin{aligned}\theta &= \lambda(1 + \exp[-i(Q/D)]) \\ &+ \exp[-2i(Q/D)] \\ &+ \exp[-3i(Q/D)] + \dots\end{aligned}\quad (2)$$

Defining the current value of the reduction in the maximum attainable cash float from any period T into the future as θ_T , then

$$\begin{aligned}\theta_T &= \lambda \exp(-iT)(1 + \exp[-i(Q/D)]) \\ &+ \exp[-2i(Q/D)] \\ &+ \exp[-3i(Q/D)] + \dots\end{aligned}\quad (3)$$

Thus on day T , the on-going average level of cash float that accrues over any previous T days beyond the initial τ days (π_T) is equal to the value of the maximum attainable cash float over T days reduced by the difference of two geometric series representing the value of the reduction in the maximum attainable cash float over T days, as follows:⁶

$$\begin{aligned}\pi_T &= Dc\tau(\exp(iT) - 1) \\ &+ \exp(iT)[\lambda(1 + \exp[-i(Q/D)]) \\ &+ \exp[-2i(Q/D)] \\ &+ \exp[-3i(Q/D)] + \dots] \\ &- \lambda \exp(-iT)(1 + \exp[-i(Q/D)]) \\ &+ \exp[-2i(Q/D)] \\ &+ \exp[-3i(Q/D)] + \dots \\ &= \left\{ Dc\tau + \left[\frac{\lambda}{1 - \exp[-i(Q/D)]} \right] \right\} \\ &\times [\exp(iT) - 1]\end{aligned}\quad (4)$$

Similarly, the contribution from sales of this product over T days equals δ_s , where:

$$\begin{aligned}\delta_s &= \int_0^T D(p - c)\exp(it) dt \\ &= D(p - c) \left[\frac{(\exp(iT) - 1)}{i} \right].\end{aligned}\quad (5)$$

Finally, denoting m as the net margin, the profits from sales of this product over T days equal π_s , as follows:

$$\begin{aligned}\pi_s &= \int_0^T Dpm \exp(it) dt \\ &= Dpm \left[\frac{(\exp(iT) - 1)}{i} \right].\end{aligned}\quad (6)$$

APPENDIX II

Deriving the Benefits From Changing a Category's Sales Mix

Consider a product group with n products where the terms for product 1 have recently been renegotiated. Subscripting for each product and expanding equations (4) and (5) from Appendix I, the total contribution from sales and float for these products before cannibalizing sales is equal to

$$\sum_{j=1}^n (\delta_{sj} + \pi_{fj})_b,$$

as follows

$$\begin{aligned}&\sum_{j=1}^n (\delta_{sj} + \pi_{fj})_b \\ &= \left[\frac{D_1(p_1 - c_1)}{i} + D_1c_1 \right. \\ &\quad \times \left[\frac{1}{i} + \tau_1 - \frac{Q_1}{D_1} \left(\frac{1}{1 - \exp[-i(Q_1/D_1)]} \right) \right] \\ &\quad + \sum_{j=2}^n \left\{ \frac{D_j(p_j - c_j)}{i} + D_jc_j \right. \\ &\quad \times \left[\frac{1}{i} + \tau_j - \frac{Q_j}{D_j} \left(\frac{1}{1 - \exp[-i(Q_j/D_j)]} \right) \right] \left. \right\} \left. \right] \\ &\quad \times (\exp(iT) - 1).\end{aligned}\quad (7)$$

Similarly, after discounting the price of product 1 by a factor α ($0 < \alpha < 1$) and cannibalizing a proportion of the sales of each of the competing products by a factor ϕ_j ($0 < \phi_j < 1$), the total contribution from these products will be equal to

$$\sum_{j=1}^n (\delta_{sj} + \pi_{fj})_a,$$

⁶The difference of two infinite geometric series is given in [18].

as follows

$$\begin{aligned} & \sum_{j=1}^n (\delta_{sj} + \pi_{fj})_a \\ &= \left[\frac{\beta_1(p_1(1-\alpha) - c_1)}{i} + \beta_1 c_1 \right. \\ & \quad \times \left[\frac{1}{i} + \tau_1 - \frac{Q_1}{\beta_1} \left(\frac{1}{1 - \exp[-i(Q_1/\beta_1)]} \right) \right] \\ & \quad + \sum_{j=2}^n \left\{ \frac{\beta_j(p_j - c_j)}{i} + \beta_j c_j \right. \\ & \quad \times \left[\frac{1}{i} + \tau_j - \frac{Q_j}{\beta_j} \left(\frac{1}{1 - \exp[-i(Q_j/\beta_j)]} \right) \right] \left. \right\} \left. \right] \\ & \quad \times (\exp[iT] - 1). \end{aligned} \quad (8)$$

where:

D = total category sales before marketing discounted product 1

$= D_1 + D_2 + \dots + D_n$

γ = growth of category sales after discounting product 1 ($\gamma > 0$)

β_j = sales of product j after marketing discounted product 1

$$\beta_1 = D_1 + \gamma D + \sum_{j=2}^n \phi_j D_j \quad (9)$$

$$\beta_j = (1 - \phi_j) D_j \quad j = 2, 3, \dots, n. \quad (10)$$

As noted by equation (9), the category sales increase as a result of the growth in sales of discounted product 1. Not only are the sales of the competing products cannibalized, but additional sales are generated by regular customers increasing their purchases and new customers from competitor stores buying discounted product 1 at this firm's stores. The benefits from the change in mix in the category's sales equal Δ , as follows:

$$\Delta = \sum_{j=1}^n (\delta_{sj} + \pi_{fj})_a - \sum_{j=1}^n (\delta_{sj} + \pi_{fj})_b. \quad (11)$$

APPENDIX III

Evaluating the Assumption of Constant Demand

The assumption of constant demand makes the analyses of the future value of cash float (π_f)

and sales (π_r and δ_s) in equations (1)–(6) from Appendix I significantly less complex and does not appreciably affect the results. Proof of the latter is provided in this Appendix in the form of simple analytical propositions and an experimental simulation. The following additional variables are used in this Appendix:

$\hat{D}_t$ = actual daily demand in day t

$\hat{\lambda}_k$ = actual value of λ for cycle k

r = number of order cycles over T days.

If the actual daily demand ($\hat{D}_t$) varies about a stationary expected value (D), then

$$\frac{\sum_{t=1}^T \hat{D}_t}{T} \rightarrow D \text{ and } \frac{\sum_{t=1}^T \frac{Q}{\hat{D}_t}}{T} \rightarrow \frac{Q}{D} \text{ for large } T. \quad (12)$$

As a result, the actual position of float relative to the maximum during each cycle ($\hat{\lambda}_k$) will also vary about the expected position in each cycle (λ) and

$$\frac{\sum_{k=1}^r \hat{\lambda}_k}{r} \rightarrow \lambda \text{ for large } r. \quad (13)$$

Hence, the estimates obtained for π_f , δ_s , and π_s from equations (1)–(6) should be reasonably accurate when the number of days (T) and cycles (r) are large and demand is stationary. If there are systematic variations in demand (such as trend or seasonality), projected demand in equations (1)–(6) can be modified accordingly. Since the purpose of this analysis is to estimate the average value of funds received over a relatively long period of time, both T and r will be large and the assumption of constant demand will not appreciably affect the results.

A simulation analysis measuring the absolute percent error between projected and actual values for float and the contribution from sales (profits from sales would behave similarly to contribution) supports this conclusion. In this simulation, the coefficient of variation (the standard deviation divided by the mean) of actual demand ($\hat{D}_t$) was normally distributed about D as an experimental factor with levels equal to 0.1, 0.2, 0.3, 0.4, and 0.5. A coefficient of variation (CV) of 0.5 represents highly variable demand. The expected number of orders (DT/Q) was defined as a second experimental

Table 1. Float results (absolute percent change from actual)

DT/Q	CV	Run 1	Run 2	Run 3	Run 4	Run 5	Cell max	Cell avg
20	0.1	0.075	0.042	0.075	0.219	0.003	0.219	0.083
20	0.2	0.044	0.332	0.305	0.230	0.022	0.332	0.187
20	0.3	0.166	0.260	0.399	0.008	0.093	0.399	0.185
20	0.4	0.256	0.161	0.235	0.347	0.738	0.738	0.347
20	0.5	0.083	0.307	0.354	0.411	0.136	0.411	0.258
30	0.1	0.038	0.166	0.161	0.305	0.105	0.305	0.155
30	0.2	0.045	0.054	0.057	0.330	0.011	0.330	0.099
30	0.3	0.014	0.014	0.082	0.293	0.058	0.293	0.092
30	0.4	0.147	0.028	0.040	0.105	0.130	0.147	0.090
30	0.5	0.466	0.012	0.168	0.450	0.086	0.466	0.236
40	0.1	0.032	0.068	0.128	0.231	0.027	0.231	0.097
40	0.2	0.132	0.064	0.150	0.388	0.114	0.388	0.170
40	0.3	0.083	0.011	0.349	0.478	0.042	0.478	0.193
40	0.4	0.400	0.025	0.131	0.166	0.210	0.400	0.186
40	0.5	0.100	0.031	0.080	0.053	0.135	0.135	0.080
50	0.1	0.013	0.066	0.017	0.106	0.060	0.106	0.052
50	0.2	0.181	0.166	0.087	0.007	0.061	0.181	0.100
50	0.3	0.020	0.146	0.048	0.149	0.110	0.149	0.095
50	0.4	0.074	0.315	0.145	0.236	0.053	0.315	0.165
50	0.5	0.102	0.181	0.113	0.235	0.225	0.235	0.171

Overall avg. = 0.152

factor with levels of 20, 30, 40, and 50. These two factors more than cover the range of grocery retail environments described by this research. Replenishment lead times were set to zero to avoid customer service and safety stock considerations from masking the effect of demand uncertainty on float and sales value estimates.

To evaluate the value of float estimates obtained with equation (4) an experiment was conducted with five replications in each of the 20 experimental cells (5 CV values times 4 levels of expected orders), for a total of 100 experimental runs. The results of this experiment are

presented in Table 1. Notice that the projected float values vary from actual by an average of only 0.152%. Furthermore, the maximum deviation between projected and actual values for all 100 runs is <1% (0.738%) and there is no apparent systematic degradation of the results over the range of values tested.

For each of the 100 runs noted above the deviation between projected and actual values for the contribution from sales was also recorded. Because the contribution from sales is not a function of the ordering cycle, this effectively resulted in 20 replications in each of the CV cells. These results, shown in Table 2, are

Table 2. Contribution results (absolute percent change from actual)

DT/Q	CV	Run 1	Run 2	Run 3	Run 4	Run 5	Cell Max	Cell Avg	Overall CV Avg.
20	0.1	0.255	0.110	0.007	0.222	0.106	0.255	0.140	0.181
20	0.2	0.199	0.146	0.018	0.099	0.093	0.199	0.111	0.096
20	0.3	0.082	0.026	0.050	0.089	0.182	0.182	0.086	0.121
20	0.4	0.199	0.154	0.245	0.211	0.274	0.274	0.217	0.170
20	0.5	0.017	0.153	0.040	0.338	0.154	0.338	0.140	0.176
30	0.1	0.227	0.276	0.268	0.058	0.227	0.276	0.211	
30	0.2	0.026	0.049	0.076	0.043	0.037	0.076	0.046	
30	0.3	0.139	0.139	0.189	0.171	0.066	0.189	0.141	
30	0.4	0.068	0.145	0.010	0.194	0.390	0.390	0.161	
30	0.5	0.104	0.269	0.064	0.360	0.116	0.360	0.183	
40	0.1	0.153	0.092	0.070	0.225	0.040	0.225	0.116	
40	0.2	0.149	0.029	0.021	0.178	0.178	0.178	0.111	
40	0.3	0.011	0.129	0.248	0.333	0.026	0.333	0.149	
40	0.4	0.250	0.160	0.043	0.234	0.024	0.250	0.142	
40	0.5	0.102	0.129	0.310	0.694	0.013	0.694	0.250	
50	0.1	0.400	0.272	0.183	0.384	0.052	0.400	0.258	
50	0.2	0.069	0.160	0.060	0.112	0.184	0.184	0.117	
50	0.3	0.107	0.089	0.206	0.132	0.008	0.206	0.108	
50	0.4	0.216	0.254	0.127	0.002	0.205	0.254	0.161	
50	0.5	0.214	0.081	0.294	0.036	0.029	0.294	0.131	

Overall avg. = 0.149

very similar to the results for float. The projected contribution varies from actual by an overall average of only 0.149% and the maximum deviation for all 100 runs is also <1% (0.694%). In addition, the overall cell averages (20 runs per cell) do not appear to be a function of the variability of demand (CV) over the range of values tested.

The results from this simulation experiment clearly show that the propositions defined by equations (12) and (13) are reasonable for the environments investigated in this research. Low estimates in some cycles are offset by high estimates in other cycles. As long as the goal is to project the value of float and sales over relatively long periods of time and there are no systematic patterns in demand, the assumption of constant demand should not significantly affect the accuracy of projections based on equations (1)–(6).

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